Gradient Guided Multiscale Feature Collaboration Networks for Few-Shot Class-Incremental Remote Sensing Scene Classification

Wuli Wang, *Member, IEEE*, Li Zhang, Sichao Fu, *Graduate Student Member, IEEE*, Peng Ren, *Senior Member, IEEE*, Guangbo Ren, Qinmu Peng, and Baodi Liu, *Member, IEEE*

***Abstract*— Few-shot class-incremental learning has recently received significant research focus in remote sensing scene classification (FSCIL-RSSC). The success of FSCIL-RSSC relies on the robustness of the feature backbone and classifiers. Existing works focus on improving classifier adaptation, but little attention is paid to the importance of backbone robustness on the recognition ability of new class samples' embeddings. Due to the large distribution shift between old and new classes, FSCIL-RSSC using high-layer (single-scale) features may not adapt flawlessly to new categories. To solve the issue, we put forward a gradient guided multiscale feature collaboration network (G-MFCN) for FSCIL-RSSC. Specifically, we introduce a parallel hierarchy strategy to simultaneously capture the multifeature discriminative information of the same sample. Then, a gradient guide block is designed to automatically pick out the optimal values of different convolution blocks for multifeature fusion. Finally, the classical feature pyramid network is introduced for multiscale fusion to obtain more obvious discriminative features of RSSC. More importantly, our proposed G-MFCN is a simple and adaptable module, which can combine any existing FSCIL frameworks to further improve the optimized classifiers' effectiveness for the FSCIL-RSSC scenario. Extensive experiments on four benchmarks demonstrate that the proposed G-MFCN achieves significant improvements in comparison to existing FSCIL-RSSC methods.**

Manuscript received 11 September 2023; revised 10 December 2023 and 21 January 2024; accepted 18 February 2024. Date of publication 23 February 2024; date of current version 4 March 2024. This work was supported in part by the Funds of the National Natural Science Foundation of China under Grant 42076189 and Grant 52075057; in part by the Key Laboratory of Land Satellite Remote Sensing Application, Ministry of Natural Resources of the People's Republic of China, under Grant KLSMNR-G202309; in part by the National Key Research and Development Program of China under Grant 2022YFF0712300; and in part by the Fundamental Research Funds for the Central Universities under Grant 22CX01004A-8 and Grant 27RA2322009. *(Corresponding authors: Wuli Wang; Sichao Fu.)*

Wuli Wang, Li Zhang, and Peng Ren are with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: wangwuli@upc.edu.cn; zhangli311415@163.com; pengren@upc.edu.cn).

Sichao Fu and Qinmu Peng are with the School of Electronic Information and Communications, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: fusichao_upc@163.com; pengqinmu@hust.edu.cn).

Guangbo Ren is with the Laboratory of Marine Physics and Remote Sensing, First Institute of Oceanography, Ministry of Natural Resources, Qingdao 266061, China (e-mail: renguangbo@126.com).

Baodi Liu is with the College of Control Science and Engineering, China University of Petroleum (East China), Qingdao 266580, China (e-mail: thu.liubaodi@gmail.com).

Digital Object Identifier 10.1109/TGRS.2024.3369083

***Index Terms*— Few-shot class-incremental learning (FSCIL), multiscale feature collaboration, remote sensing scene classification (RSSC).**

I. Introduction

REMOTE sensing scene classification (RSSC) is a kind of technology for ground object type classification by analyzing various characteristics and attributes in remote sensing images, which can provide accurate information support for geological exploration, environmental monitoring, road planning, and other fields [1], [2]. Compared with early methods of RSSC based on traditional machine learning [3], [4], the application of deep learning in RSSC has received significant attention due to their strong ability in feature representation and classification decision-making [5], [6], [7]. However, deep learning methods require enough labeled samples for training to achieve superior performance, which poses a considerable challenge for most practical applications due to factors such as terrain, weather, and equipment.

To address the problem of sample shortage, researchers have designed a novel method called few-shot learning (FSL), which can learn new knowledge from scarce labeled samples based on prior knowledge for accurate classification [8], [9]. For instance, Li et al. [10] combined the matching network and attention mechanism for few-shot RSSC, which achieves marvelous results and also becomes a benchmark. Yuan and Huang [11] proposed a deepEMD network with a multiattention mechanism, which effectively exploits delicate local features while mitigating interferences introduced by the complex background. Han et al. [12] designed an MVFF method to alleviate the issue of distribution deviation in the extracted features when using a single model. Although the above FSL methods have significantly improved the performance of RSSC under scarce labeled samples, the existing FSL methods focus more on new class samples and even retrain the entire model during continuous testing, leading to catastrophic forgetting knowledge of old class.

To solve the catastrophic forgetting issue of old classes in FSL, a novel few-shot class-incremental learning (FSCIL) method was proposed, which integrates new data with previous knowledge, gradually introduces new categories, and maintains the memory of old knowledge [13]. In the past several

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

years, many FSCIL methods have been proposed [14], [15] and achieved state-of-the-art performance in various domains. Song et al. [16] proposed a novel semantically aware virtual contrast (SAVC) framework to enhance FSCIL by learning recognition and contrast in virtual space. It retains enough positions for the new generalization and achieves a multisemantic aggregation reasoning effect. Pan et al. [17] proposed a self-supervised learning (SSL) and knowledge distillation (KD) framework to improve feature extraction in low-shot classes for fine-grained visual classification (FGVC), leading to enhanced classification in SSFE-Net.

Recently, some researchers have introduced FSCIL methods into the field of remote sensing, which also provides novel avenues for RSSC (FSCIL-RSSC). For instance, Liu et al. [18] proposed an incremental learning framework based on open set recognition (ILOSR) for RSSC in open and dynamic environments, which can identify unknown classes from data streams and gradually learn these new classes. Zeng et al. [19] proposed an FSCIL-RSSC model based on feature adaptation and prototype continuous optimization to improve the recognition ability of the incremental phase. Zhu et al. [20] put forward an FSCIL method for precise classification of remote sensing images, which can relieve the serious interclass confusion and mitigate the catastrophic forgetting. Tai et al. [21] proposed the Mine-Distill Prototypes algorithm, which seeks to minimize the influence of unsuitable channels on classification and focuses on solving catastrophic forgetting problems by learning the corresponding dimensions of the new prototype.

However, most existing FSCIL-RSSC works tend to design more complex strategies to improve the fast adaptation of the classifiers on the new classes, and little effort is concentrated on using the multiscale features extracted from pretrained feature backbone reasonably to enhance the discrimination of new class samples' embeddings. The performance of the classifier depends on the robustness of the pretrained backbone, and classifier-based improvement methods can only target specific models and specific tasks. In the real remote sensing scenario, there is not a large amount of sample data available for use. Owing to the large class distribution gap between the base and incremental sessions, and data scarcity on the incremental sessions, this may make it difficult for models to distinguish between old and new categories. Therefore, how to pretrain a generality and adaptability backbone network and how to use limited data to enhance its generalization ability to obtain more discriminative features are important issues to be addressed in FSCIL-RSSC.

In this article, we propose a gradient guided multiscale feature collaboration network (G-MFCN) for FSCIL-RSSC. Our G-MFCN can offset single-scale deficiencies and incorporate multiscale information to capture more precise and comprehensive features, and then enhance the discriminative feature representation ability of the existing FSCIL-RSSC model. Compared with the current FSCIL-RSSC methods, we concentrate on enhancing the resilience of feature extractors trained on the base classes instead of improving the fast adaptation of the classifiers on the new classes. By emphasizing the elasticity of the feature extractor trained on the base class, the model can better learn the common features, which helps improve the transfer learning effect between different tasks. In this way, the model is better able to adapt even if the data distribution of the new category varies from the base category. In addition, by improving the robustness of the feature extractor, the model can better maintain the memory of previous knowledge and reduce the forgetting problem.

Specifically, we use a parallel hierarchy strategy to simultaneously capture the multiscale discriminative information of the same sample and fuse the features of the four stages with two branches at the same scale. Then, a gradient guided block module is designed to calculate the importance score of each convolution layer for the feature extraction situation of different pictures and then obtain the importance of each convolution layer relative to the feature extraction of the whole dataset. This process allows for selecting the optimal multiscale fusion layers for diverse datasets, significantly enhancing the scene classification accuracy. Finally, the classic feature pyramid network is introduced for adaptively fusing multiscale features to achieve more discriminative representations. What is particularly attractive is that our G-MFCN can be flexibly integrated with the current FSCIL frameworks to generate more effective feature embeddings for incremental classes. The main contributions are presented as follows.

1) We put forward a novel G-MFCN framework for FSCIL-RSSC, which aims to improve the robustness of the pretrained backbone in enhancing the feature extraction of new class samples. As far as we know, we are the first to reveal and address the importance of the pretrained backbone robustness in the FSCIL-RSSC scenario.
2) To obtain more comprehensive information from limited data, a parallel hierarchical strategy is designed to perform multiple feature fusion, and a feature pyramid structure is introduced for multiscale feature fusion. To overcome the constraints of conventional multiscale fusion methods, a gradient guidance module is proposed for evaluating each convolution layer's contribution which aims to pick out the optimal of the corresponding multiscale semantic features automatically.
3) Our G-MFCN is also a universal and flexible module, which can combine any existing FSCIL frameworks to further enhance the optimized classifiers' generalization in the FSCIL-RSSC scenario.

II. Related Work

In this segment, some related works about FSCIL and multiscale learning will be introduced in detail.

A. Few-Shot Class-Incremental Learning

In previous works, incremental learning and FSL have been extensively studied [22], [23]. Incremental learning enables system to gradually introduce new data to update learning on the basis of already learning part of the data, without retraining entire model. This method allows for continuous learning in real-time applications, continuously enhancing the performance of the model. Recently, class-incremental learning (CIL) has attracted more attention, Li and Hoiem [24] proposed a novel network structure and training strategy that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

only uses new task instances and saves the existing task parameters in the original network, which effectively mitigates forgetting problem of old category knowledge.

Due to the ability of FSL to solve the problem of sample shortage in practical applications through rapid learning, transfer learning, and model generalization with only a few samples, the combination of FSL and CIL (FSCIL) has become a research hotspot over the past years. Cheraghian et al. [25] introduced a method for KD with semantic awareness that combined previously learned knowledge with the semantic information of new tasks to improve generalization and adaptability of FSCIL. Zhu et al. [26] used quantized vectors in deep embedded space to coordinate old knowledge and acquire new knowledge effectively, which greatly relieves the overfitting and catastrophic forgetting problems in FSCIL. Zhang et al. [14] put forward a FSCIL approach based on a decoupling strategy that decoupled feature extractor from classifier and only updated the classifier in each incremental stage. Zhou et al. [15] proposed a learning compatibility training (FACT) for FSCIL that can solve the impact of the poor work efficiency of the previous model on the latter model. FACT adeptly integrates new classes while maintaining compatibility with old ones, exhibiting robustness against negligence. In recent years, there has been a progressive integration of FSCIL into RSSC. Zeng et al. [19] put forward an FSCIL-RSSC model using feature adaptation and continuous prototype optimization to improve recognition in the incremental phase.

Although the above methods have made some achievements, most existing FSCIL-RSSC methods directly use the high-layer (single-scale) semantic features for classification, which often ignore the details of shallow features and affect the performance of RSSC. Thus, there is considerable scope for incorporating multiscale learning in FSCIL-RSSC in the future.

B. Multiscale Learning

Integrating features at different scales is crucial for improving performance in various deep learning tasks such as image classification and object detection. Low-level features possess higher resolution and more detailed information but lack semantic context. While high-level features provide stronger semantic information but have lower resolution and limited perception of details. Efficiently integrating the advantages of both is critical for improving classification models. Compared with ordinary optical images, remote sensing images have characteristics such as diverse land cover, blurry boundaries of objects, and rich semantic information. Therefore, effectively fusing multiscale information to obtain more discriminative features is a pressing issue that needs addressing in RSSC. Lin et al. [27] developed a top–down architecture with lateral connections for the fusion of features at different scales. Liu et al. [28] proposed a bottom-up data augmentation on the basis of feature pyramids, which can better preserve low-level features. Li et al. [29] put forward a bidirectional fusion network structure, which takes into account the different contributions of features and introduces weights for calculation during the fusion stage. Fu et al. [30] built a dual-branch network, which harnesses attention mechanism to select and fuse features adaptively.

However, multiscale fusion has not been fully used in the field of FSCIL-RSSC, and the existing methods lack a criterion for quantitatively analyzing the performance of each layer of the backbone network. In this article, we propose a gradient guided multiscale fusion method for FSCIL-RSSC, aiming to train a more robust feature extractor for the incremental stage, thereby improving classification performance.

Algorithm 1 Simplified Training Procedure of G-MFCN. y_q and $\hat{y}_q$ Indicate the Ground-Truth Label and the Network Predictions; $L(\cdot)$ is the Cross-Entropy Loss Function

Input: Base class session D_0
while not done **do**
  Get multiscale feature R_k and L_k on the two parallel feature extractors via (1) and (2)
  Add GMF to calculate importance score for each layer of ResNet-18 via (4)
  Get the feature R'_k corresponding to the optimal layer in each convolution block
  Get F_k with fusing the R'_k and L_k via (5)
  Introduce FPN to accomplish multiscale fusion and get the fusion feature M_k via (6)
  Compute loss with $L(y_q, \hat{y}_q)$ and optimize the feature extractor parameters
end while
return the optimal feature backbone trained on D_0

III. Method

Owning to interclass similarity of a high level or degree and heavy intraclass difference in remote sensing images, using a single feature extractor may lead to issues of insufficient feature representation and single using the high-layer (single-scale) semantic features usually cannot adapt to the new classes flawlessly because of the large distribution shift between old and new class samples. Therefore, we put forward a G-MFCN for achieving the better performance of RSSC, as shown in Fig. 1. First, G-MFCN captures two distinct sets of features from two different convolutional neural networks (CNNs) by a parallel hierarchy strategy. Then, a gradient guide block is designed to automatically pick out the optimal values of different convolutional blocks for multifeature fusion. Finally, the classical feature pyramid network is introduced for multiscale fusion to obtain more obvious discriminative features of RSSC. The pipeline of our method can be seen in Algorithm 1.

A. Problem Formulation

The goal of FSCIL is to create a deep learning algorithm capable of acquiring new knowledge from a limited set of new training examples, while simultaneously preserving previously learned knowledge of old classes without experiencing forgetting. In this article, we use $\{D^0, D^1, \ldots, D^i\}$ to present the training sets of different learning sessions, and the corresponding label space of dataset D^i is indicated by C^i $(i \geq 0)$. Here,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

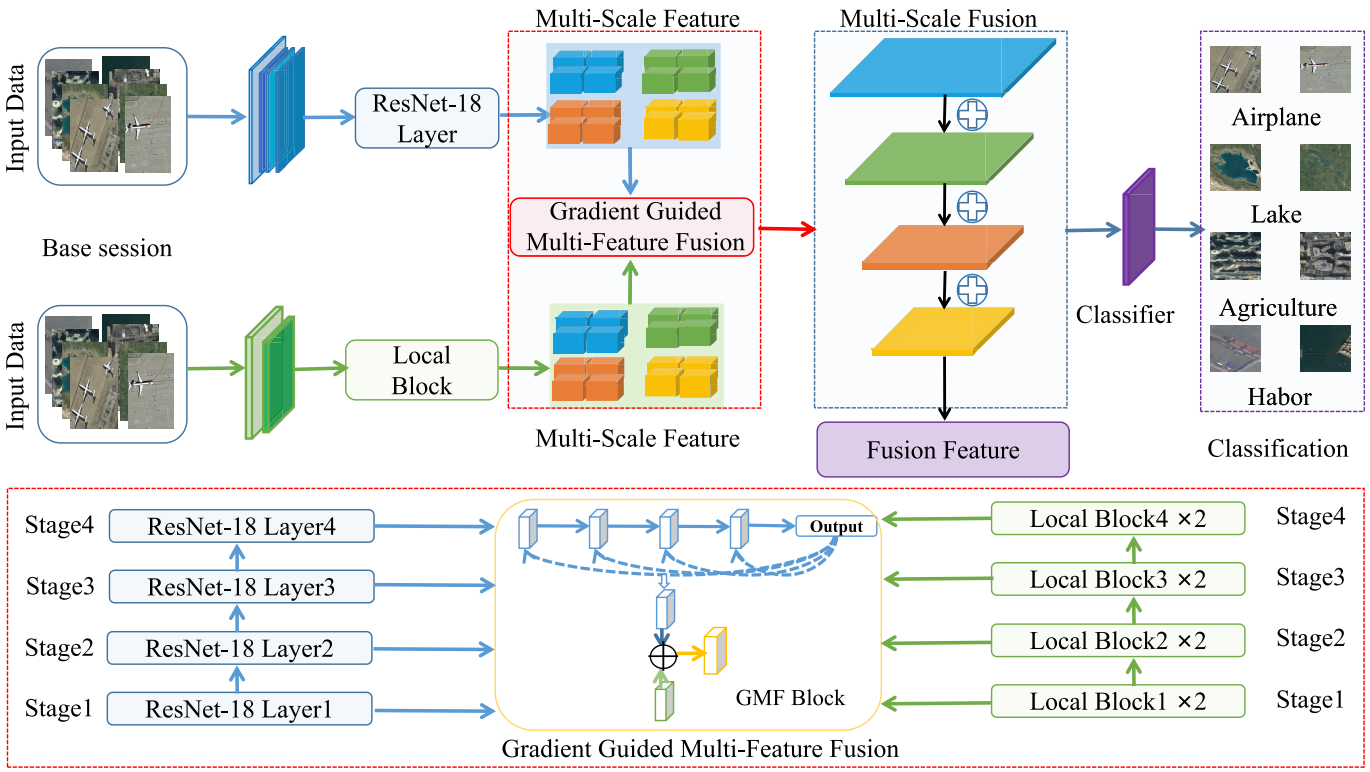


Fig. 1. Basic framework of the G-MFCN. The model network conducts two parallel feature extractors, the GMF and the multiscale fusion block. Regarding the incremental learning task process, we have specifically shown in Fig. 6.

there are no overlapped classes between different datasets, $\forall i, j$ and $i \neq j$, $C_i \cap C_j = \emptyset$. For G-MFCN, the feature extraction network structure is trained using the training data from the initial session D^0. Then, the pretrained feature collaboration network is adapted to the new classes in incremental sessions $D^i (i > 0)$. Generally, the dataset in the first session D^0 has sufficient data for training, defined as the base training set. The dataset in incremental sessions D^i $(i > 0)$ has limited data for training, referred to as a training set with N-way and K-shot. For instance, in the UCMerced-LandUse benchmark dataset, the base session consists of 12 classes, with each class having 60 training images. However, in each subsequent incremental session, only three classes are used for training, with a restriction of five images per class.

B. Multiscale Feature Extraction

The performance of FSCIL-RSSC mainly relies on the base class feature extractor's resilience and the new class classifier's adaptability. However, the existing methods mainly prioritize the development of intricate structures to enhance the adaptability of classifiers, while neglecting the potential enhancement of feature extractors through the utilization of multiscale features. To achieve comprehensively multiscale features from remote sensing scene, we design a parallel multilevel structure with double branches. We choose residual network (ResNet-18) as the backbone of our framework and introduce an improved deeply separable network structure [31], [32] to fuse with it. The parallel hierarchy strategy can simultaneously capture the multiscale discriminative information from four stages. It can not only better concentrate on the local information of the image but also compensate for the deficiency of individual feature extractors.

ResNet-18 comprises four convolutional layers with residuals and a layer for average pooling, each as shown in Fig. 2(a), and the output of each convolution block is represented as follows:

$$R_k = F(X_k) + X_k \tag{1}$$

where F represents the nonlinear transformation function within the residual block, typically made up of convolution layers, batch normalization layers, and activation functions. X_k indicates the input feature to the ith residual block. $+$ is the additive operation at the element level, and R_k stands for output features.

The local block adopts the structure of deepwise separable convolution effectively which reduces the floating point operations (FLOPs) of the network. In addition, it sets distinct expansion rates in different periods so that the deep model is more efficient in image analysis and calculation while improving accuracy and robustness. Each is shown in Fig. 2(b), and the outcome of each block can be depicted as follows:

$$L_k = f^{d1\times1}\left(\mathrm{LN}\left(f^{d3\times3}(L_{k-1})\right)\right) + L_{k-1} \tag{2}$$

where L_k denotes the output from the local feature block. $f^{dn\times n}$ signifies the depthwise convolution process using an $n \times n$ kernel size.

In addition, considering the insufficient samples in the training set, we expand samples by random tailoring, random scale, and random level of automatic data enhancement. Then, we use the newly generated image together with the original

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

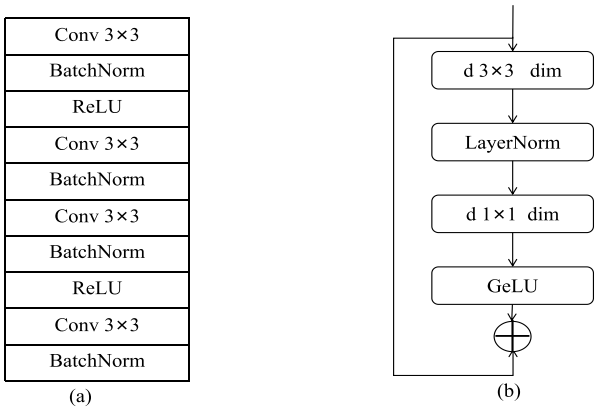


Fig. 2. Structure of (a) ResNet-18 layer and (b) local block.

image to train our fusion network, and the network optimization loss function is represented as follows:

$$L = (1 - \lambda) \cdot L(I) + \lambda \cdot L(R) \tag{3}$$

where L denotes the total loss, $L(I)$ indicates the loss of original images, $L(R)$ stands for the loss of the generated images, and λ represents a hyperparameter used for balancing the types of loss.

C. Gradient Guided Multiscale Feature Collaboration

ResNet-18 is a typical feature-extracted CNN composed of 18 layers and four convolution blocks from shallow to deep. To more intuitively understand each layer of extraction features, we use gradient-weighted class activation mapping (Grad-CAM) to present features extracted from the 16 convolution layers, as depicted in Fig. 3.

Using heat maps to show the feature extraction of each layer of ResNet-18 helps us understand the hierarchical feature extraction ability and information flow of the network. Specifically, the heat map can reflect the importance and activation of the feature maps of each layer. According to the heat map, we can discover which layers are more critical for the feature extraction of different categories, and which layers play a significant role in enhancing network performance.

Most existing fusion methods based on ResNet combine the last layer output of different convolutional blocks, but the other layers of each convolution block may produce better results for different categories. Therefore, it is essential to prejudge the image before feature extraction and select the optimal with the best effect for fusion. In addition, the feature extraction situation of each layer is also different for various images, and it is difficult to show the feature extraction of each image for the whole dataset using heat maps. So we design a collaboration network based on gradient guidance, and the operation of each convolution block is shown in Fig. 4.

In the gradient guidance section, to make a qualitative judgment, we calculate the significance score of each layer of ResNet-18 by gradient average pooling. Specifically, 1) for each input map, we propagate its forward to the last layer of each convolution block of ResNet-18; 2) through the backpropagation algorithm, we calculate the gradient of each output unit for each output feature map and calculate the average gradient; and (c) assigning the importance score of each feature map based on the average gradient size of its values, the importance score of each layer of the convolutional block can be obtained. The process is depicted as follows:

$$S_i = S_{i-1} + \text{mean}(\text{abs}(\text{mean}(\nabla(y_i, w_i), \text{dim} = [2, 3]))) \tag{4}$$

where S_i is the score of each image in the layer, y_i represents the output of each convolution block, and w_i represents the weight parameters of the ith convolutional layer in the ith residual block. dim $= [2, 3]$ stands for average operations on the second and third dimensions; by averaging over the spatial dimension, we can obtain the average sensitivity of each weight parameter over the entire feature map, thus more comprehensively assessing the importance of each layer to the network output.

We use the UCMerced-LandUse dataset as an example and present the importance score of features in the entire dataset as a histogram which is shown in Fig. 5.

The first layer of each convolution block extracts the features of the input data, which provides important information for the subsequent level of feature extraction. Because the information extracted by the layer will be repeatedly used in the following layers, the score of the layer tends to be high. However, the first layer usually extracts low-level local features that have little difference for objects of different categories. Moreover, the convolution operation in the first layer usually has more parameters, which increases the risk of overfitting. Thus, we do not consider the first layer score of each convolution block and choose the several layers with the highest score in the remaining layers for fusion. For remote sensing image X_k, the fusion feature through the proposed network can be determined by the following equation:

$$F_k = \{T_i[R_i(X_k)] \oplus L_i(X_k)\}_{i=1}^{4} \tag{5}$$

where F_k represents the multifeature fusion of sample X_k, $R_i[\cdot]$ indicates features generated by the ResNet-18 layer at the ith stage, $L_i[\cdot]$ stands for the features generated by the local block at the ith stage, $\oplus$ is the elementwise sum, and $T_i[\cdot]$ denotes the ith GGF block.

The aforementioned feature extractors fall into four stages, each extracting features at different scales from shallow to deep layers. With the proposed gradient guided multifeature fusion (GMF) module, we can obtain a more comprehensive set of multiscale features combining the two branches. In current FSCIL-RSSC approaches, the focus is primarily on using high-level features for the ultimate classification task. However, it is important to acknowledge that the neural network's low-level and mid-level features also carry significant detailed information. Unfortunately, as the number of convolution layers increases, these lower level features may become diluted or even vanish. Therefore, incorporating multiscale collaboration can be advantageous in capturing a more comprehensive range of information. In addition, considering the need for consistent channels in multiscale feature fusion, we adopt the fusion method of feature pyramid, combining the upsampled high-level semantic features with the low-level

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

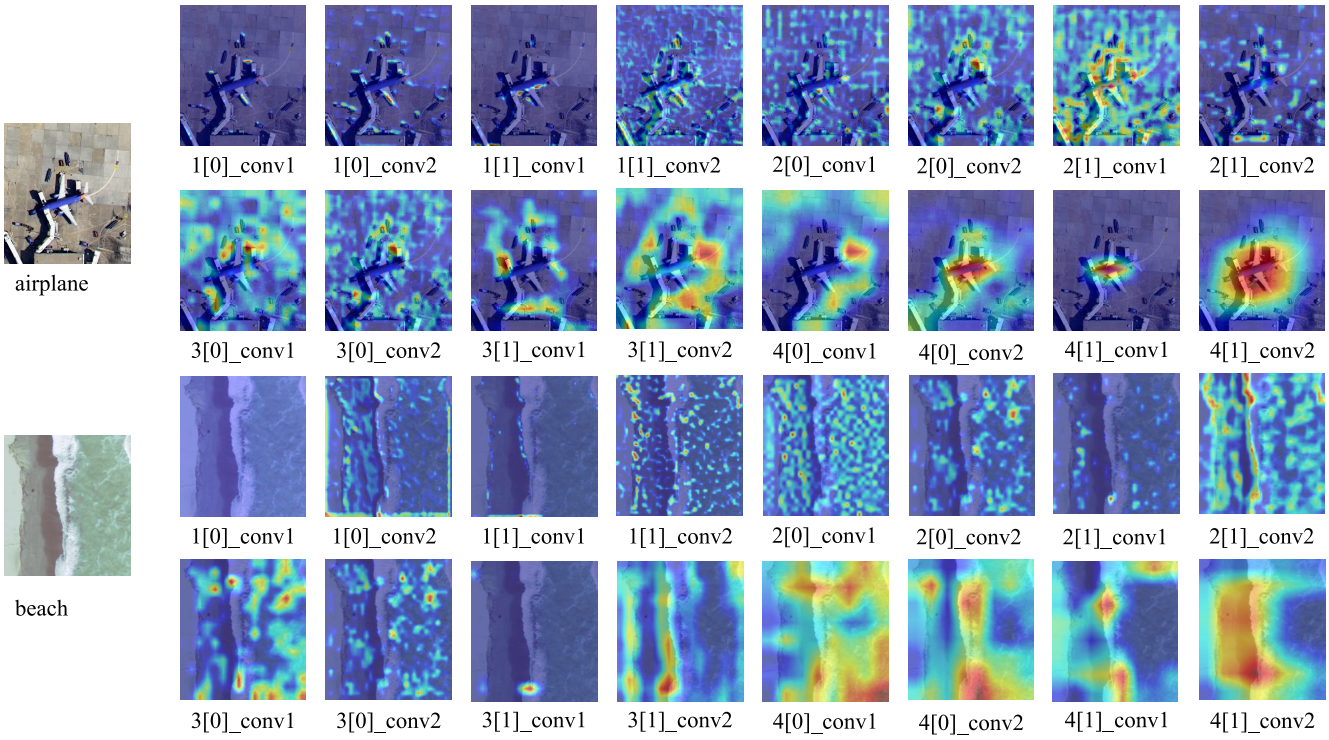


Fig. 3. Heat map visualization of each layer of ResNet-18 on the UC Merced dataset. For example, `1[0]_conv1` represents the first convolution layer of the first convolution block.

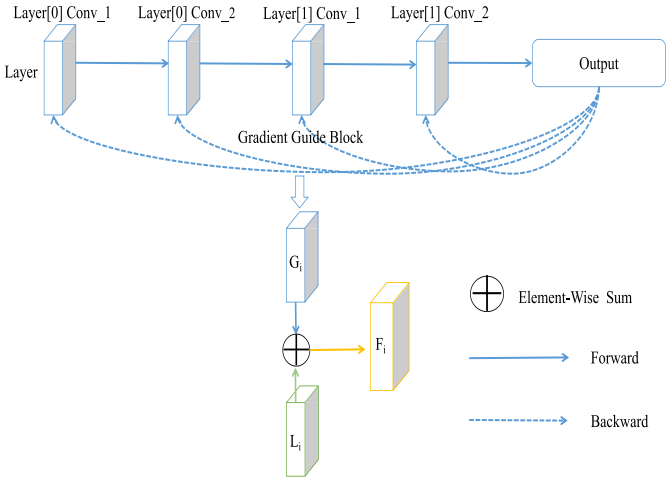


Fig. 4. Structure of the GMF. The gradient block can judge each layer of ResNet-18 and select the optimal layer for fusion.

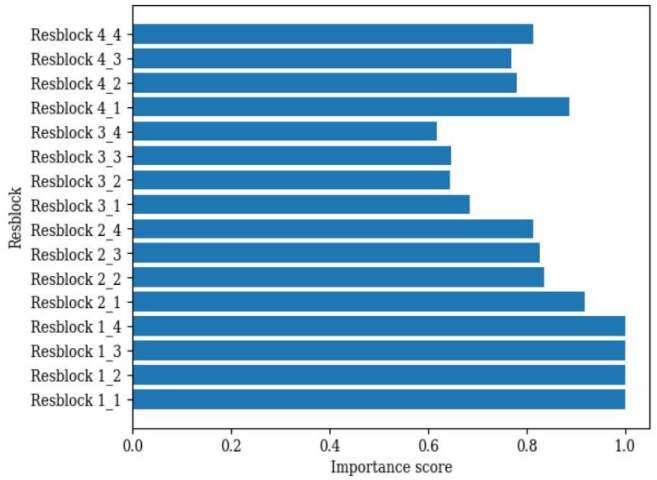


Fig. 5. Importance score for the layer of each convolutional block. The average gradient magnitude of each value on each feature map can be used as the importance score for that feature map, and then the importance scores for each layer can be obtained.

localization and detail features. The process is defined as follows:

$$M_k = F_4 \oplus \mathrm{Lat}(F_3) \oplus \mathrm{Lat}(F_2) \oplus \mathrm{Lat}(F_1) \tag{6}$$

where M_k represents the multiscale fusion features, F_K indicates features from the GMF, $\oplus$ is the elementwise sum, and Lat[·] denotes the latent layer operation that transforms features through a convolution layer to match dimensions and feature spaces.

D. Combine With Existing FSCIL Architectures

G-MFCN is a straightforward and independent module that can seamlessly integrate with current FSCIL architectures to accurately predict the classes in incremental sessions. In this section, G-MFCN is combined with the two advanced FSCIL approaches, CEC [14] and FACT [15]. We found that the module can effectively be embedded into the training tasks of feature extraction and incremental sessions, training better classifiers. Fig. 6 shows the interaction and functionality of our module with other FSCIL architectures during incremental learning sessions $D_i (i > 0)$ [33]. The experimental results of Section IV verify that the performances of the existing FSCIL architectures have been significantly improved by combining the proposed G-MFCN module with it.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

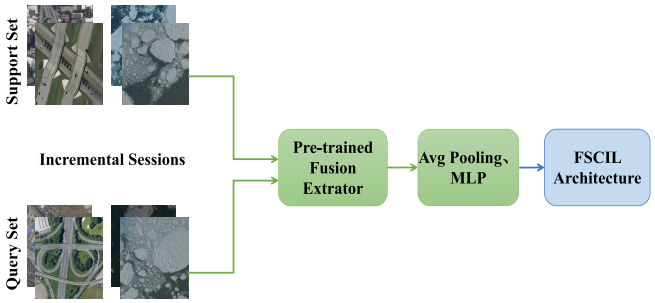


Fig. 6. Incremental inference of the current FSCIL architectures with the proposed G-MFCN module on D_i $(i > 0)$.

IV. Experiment

A. Datasets

We assess the effectiveness of G-MFCN approach on three remote sensing datasets: NWPU-RESISC45 [34], UCMerced-LandUse [35], and SIRI-WHU [2]. We select three datasets with different numbers of categories, which also proves the generality of our proposed method. In addition, since the current remote sensing dataset is not directly used in FSCIL, we complete the incremental division of the three datasets according to the format of cub200 dataset, realizing the migration of FSCIL from ordinary optical dataset to remote sensing dataset.

NWPU-RESISC45 is a dataset of 45 RSSC categories, which contains 31 500 images in total, 700 images per class. The dimension of each image is 256 × 256 × 3. Here, these 45 classes of NWPU-RESISC45 are split into 25 base classes and 20 incremental classes that are subdivided into four sessions. For the base class part, the training and testing sets are divided in a 3:1 ratio. For the incremental sessions, the training set for each session is generated by randomly choosing five training samples from each class, while maintaining initial proportion for the testing set. The UCMerced-LandUse dataset is the publicly available dataset widely used for ground object classification and image recognition. The dataset, created by researchers at the University of California, Merced (UC Merced), contains 2100 high-resolution aerial images from 21 categories, each with a size of 256 × 256. The SIRI-WHU dataset, consisting of 2400 images, was collected by Wuhan University's remote sensing team. The dataset is categorized into 12 scenes, and each image has a size of 200 × 200 pixels.

In our experiments, we divide the UCMerced-LandUse dataset into 12 base classes and nine incremental classes that are further subdivided into three sessions. We divide the SIRI-WHU dataset into six base classes and six incremental classes that are also subdivided into three sessions. For both the datasets, in the base session, we allocate a training and test set ratio of 6:4. For the incremental sessions, we adopt the same dataset splitting approach for the NWPU-RESISC45 dataset.

B. Experimental Setting

Our module is designed to be a versatile component that can integrate with the existing FSCIL methods. In this article, we choose two most classic FSCIL architectures CEC and FACT as baseline models and choose ResNet-18 as backbone of the feature extractor. The method is developed using the PyTorch framework and pretrained on a server equipped with a Tesla V100 GPU. During training, SGD is used for optimizing the model. The learning rate is set at 0.01, the batch size is set at 128, and the epoch is set to 100. Other parameters are chosen based on the specific dataset and the baseline model, following the experimental settings of CEC and FACT. At the same time, we use the data automatic enhancement method mentioned by FACT to preprocess the data. We assess the model with test set D_i after each test and report accuracy of TOP-1.

C. Comparison With Existing Methods

Most of the existing FSCIL methods are used in traditional public datasets and there are fewer methods used in the field of RSSC. To better evaluate the G-MFCN, we use some current advanced FSCIL architectures on ordinary optical images for comparison including SPPR [36], CEC [14], FACT [15], C-FSCIL [37], PCOFA [19], SSFE [17], and SAVC [16]. To accurately compare the effectiveness of G-MFCN, we introduce the average accuracy rate Average $= (1/N)\sum_i A_i$ of all the sessions to estimate the model. In Table I, we show the results of the comparison experiments, and it is clear that G-MFCN achieves the best results on the NWPU-RESISC45 dataset. More importantly, the G-MFCN model can be combined with the existing baseline structure to achieve better results. Specifically, our method is improved by 2.775% compared with the CEC baseline and by 2.815% compared with the FACT baseline. In addition, G-MFCN not only achieves improvements in average accuracy but also shows significant improvements in adaptive incremental session accuracy. This indicates that our G-MFCN model can better adapt to new categories and improve classification performance during the incremental sessions.

The experimental comparisons of UCMerced-LandUse and SIRI-WHU are shown in Tables II and III, respectively. Because there are few methods used in the RSSC and we lack the contract methods, in these two datasets, we only show the results comparing G-MFCN with the three baseline structures and two state-of-the-art methods in turn. We will provide more experimental results with FSCIL for the ordinary optical images in the Appendix to reflect the completeness of the experiments we have done. Auto represents the automatic search data augmentation method autoaugment proposed by Google, which is referred to as auto by us. Taking the baseline CEC + auto as an example, our proposed CEC + auto + G-MFCN achieves an improvement of 2.310% and 1.852%, respectively, on the two datasets.

The above results validate the effectiveness of our module with three baseline structures. The proposed G-MFCN module can effectively use multiscale information obtained by the two feature extractors, which can comprehensively fuse shallow features with deep features and improve classification accuracy. In addition, to further validate the generality of the G-MFCN module, we use VGG-16 instead of ResNet-18 as the backbone of the network. We select CEC and CEC + auto

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

TABLE I
ACCURACY OF DIFFERENT METHODS ON THE NWPU-RESISC45 DATASET

Method	Session 1	Session 2	Session 3	Session 4	Session 5	Average	Relative Improvement
Liner	92.49	14.39	14.05	13.00	12.78	29.34	+49.42
SPPR (CVPR 2021) [36]	94.18	83.02	73.45	66.57	59.37	75.32	+3.45
CEC (CVPR 2021) [14]	91.81	75.19	74.15	71.56	67.23	75.99	+2.77
FACT (CVPR 2022) [15]	92.99	77.58	73.40	69.77	66.00	75.95	+2.81
C-FSCIL (CVPR 2022) [37]	94.12	81.18	72.76	65.55	59.20	74.56	+4.20
PCOFA (ICUS 2022) [19]	94.52	83.68	75.44	67.88	61.30	76.56	+2.20
CEC+auto (TPAMI 2023) [38]	93.14	77.98	74.67	72.65	69.28	77.54	+1.22
SSFE (WACV 2023) [17]	92.70	79.40	74.76	72.89	70.26	78.00	+0.76
SAVC (CVPR 2023) [16]	94.37	81.96	74.60	71.70	66.38	77.82	+0.94
CEC+G-MFCN	92.40	80.26	74.55	72.65	67.53	77.48	+1.28
FACT+G-MFCN	93.00	80.29	75.69	72.85	68.60	78.06	+0.69
CEC+auto+G-MFCN	93.25	80.51	76.19	73.74	70.11	78.76	

TABLE II
ACCURACY OF DIFFERENT METHODS ON THE UCMERCED-LANDUSE DATASET

Method	Session 1	Session 2	Session 3	Session 4	Average	Relative Improvement
CEC (CVPR 2021) [14]	91.95	81.71	75.47	66.87	79.00	+6.45
FACT (CVPR 2022) [15]	94.70	88.14	75.96	69.99	82.19	+3.25
CEC+auto (TPAMI 2023) [38]	95.30	88.00	78.55	70.72	83.14	+2.31
SSFE (WACV 2023) [17]	97.53	82.11	75.49	72.99	82.03	+3.43
SAVC (CVPR 2023) [16]	98.25	85.16	73.09	73.29	82.44	+3.01
CEC+G-MFCN	93.20	79.42	75.84	71.79	80.06	+5.38
FACT+G-MFCN	96.70	86.85	78.87	72.16	83.65	+1.81
CEC+auto+G-MFCN	97.20	88.57	78.34	77.71	85.46	

TABLE III
ACCURACY OF DIFFERENT METHODS ON THE SIRI-WHU DATASET

Method	Session 1	Session 2	Session 3	Session 4	Average	Relative Improvement
CEC (CVPR 2021) [14]	98.95	86.71	79.87	72.30	84.46	+2.09
FACT (CVPR 2022) [15]	99.80	87.21	79.63	72.65	84.82	+1.73
CEC+auto (TPAMI 2023) [38]	98.80	87.00	80.00	73.00	84.70	+1.85
SSFE (WACV 2023) [17]	99.41	87.50	73.77	74.12	83.70	+2.85
SAVC (CVPR 2023) [16]	98.70	85.86	81.36	72.40	84.58	+1.97
CEC+G-MFCN	99.20	87.93	79.93	74.33	85.35	+1.21
FACT+G-MFCN	99.80	88.29	80.25	73.24	85.39	+1.16
CEC+auto+G-MFCN	99.40	88.50	81.88	76.43	86.55	

TABLE IV
ACCURACY OF DIFFERENT METHODS WITH VGG16 AS THE BACKBONE ON THE NWPU-RESISC45 DATASET

Method	Session 1	Session 2	Session 3	Session 4	Session 5	Average	Relative Improvement
CEC (CVPR 2021) [14]	91.04	76.06	71.69	68.85	64.70	74.45	+2.74
CEC+auto (TPAMI 2023) [38]	91.33	77.75	73.23	69.97	67.18	75.89	+1.30
CEC+G-MFCN	90.96	77.92	74.43	71.76	67.53	76.52	+0.67
CEC+auto+G-MFCN	91.70	79.69	74.11	72.43	68.53	77.19	

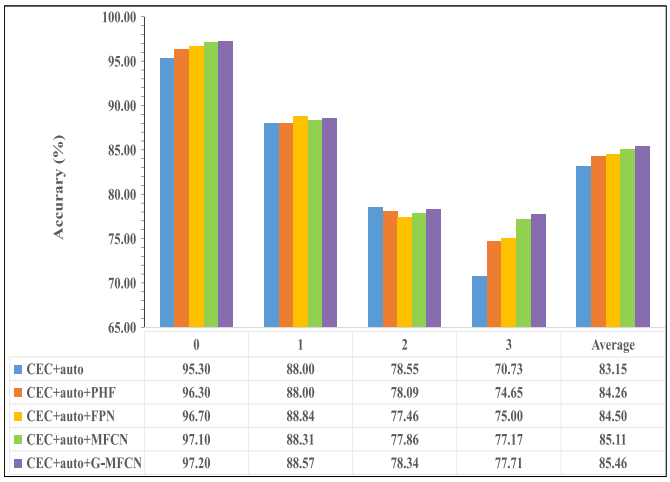
as the baseline structures and test them on NWPU-RESISC45, and Table IV lists the comparison results. Specifically, our method is improved by 2.74% compared with the CEC and by 1.30% compared with the CEC + auto. In summary, our proposed G-MFCN can be used not only with different baseline structures but also with different backbone networks.

D. Ablation Studies

In this section, the different ablation experiments we conducted, respectively, validate that the proposed multiscale fusion module and gradient guidance module are effective. We choose the UCMerced-LandUse dataset and CEC + auto as the baseline structure to perform the comparative experiments. The ablation experiment mainly consists of four parts, and the comparison of the experimental results is shown in Fig. 7.

1) *CEC + auto + PHF:* A parallel hierarchy structure is introduced into the baseline model and fused with multiple features. In this part, it is defined as PHF. It can be seen that the accuracy of this module and the result is increased by 1.11% on the CEC + auto.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.



	0	1	2	3	Average
CEC+auto	95.30	88.00	78.55	70.73	83.15
CEC+auto+PHF	96.30	88.00	78.09	74.65	84.26
CEC+auto+FPN	96.70	88.84	77.46	75.00	84.50
CEC+auto+MFCN	97.10	88.31	77.86	77.17	85.11
CEC+auto+G-MFCN	97.20	88.57	78.34	77.71	85.46

Fig. 7. Comparisons of different scales of feature and different structure of the fusion network on the UCMerced-LandUse dataset.

TABLE V
ABLATION ANALYSIS OF DIFFERENT MODULE INTEGRATION

Method	(1)	(2)	(3)	(4)	(5)
CEC+auto	√	√	√	√	√
PHF		√		√	√
FPN			√	√	√
GGM					√
Average	83.15	84.26	84.50	85.11	85.46

2) *CEC + auto + FPN:* The feature pyramid structure is introduced into the baseline model, and the features extracted from different convolution blocks of ResNet-18 are fused from top to bottom, and it is defined as FPN in this part. It can be seen that the accuracy of this module is improved by 1.35% on the CEC + auto.
3) *CEC + auto + MFCN:* We combine multifeature fusion and multiscale fusion, which can not only focus on different types of features but also have different information about both deep and shallow features. We newly define it as MFCN. As can be seen, the MFCN method is improved by 1.96% over the CEC + auto, 0.85% over CEC + auto + PHF, and 0.61% over CEC + auto + FPN.
4) *CEC + auto + G-MFCN:* Before feature fusion, we select the optimal layer of each module through the gradient guidance module (GGM) for fusion, and the result is increased by 0.355% on the basis of MFCN.

To more intuitively show the effectiveness of each module, all the above results are listed in detail in Table V. Through the above ablation experiments, we can see that our proposed multiscale fusion network can improve the classification accuracy of remote sensing scenes.

E. Parameters Sensitivity

1) Incremental Shot: During the test process, the accuracy of our proposed G-MFCN model in different situations can be demonstrated by the parameters used in the test conditions. To investigate the effect of the test condition on FSCIL, we maintain the same testing approach as the benchmark setting while varying the number of shots from {1, 3, 5, 7, 9, 11}. Then, we compare the accuracy of the incremental sessions on UCMerced-LandUse and NWPU-RESISC45, as shown in Fig. 8.

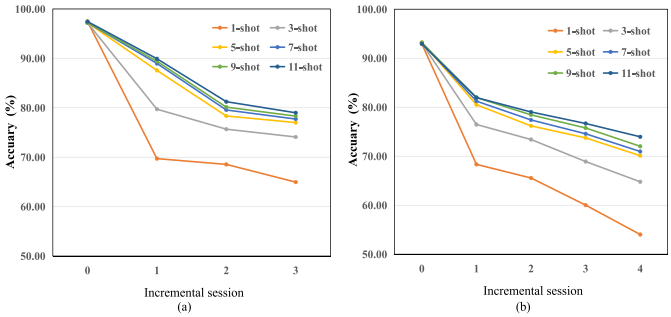


Fig. 8. Comparison of different shots for incremental learning. The accuracy of the incremental sessions is reported in (a) UCMerced-LandUse and (b) NWPU-RESISC45 for comparison.

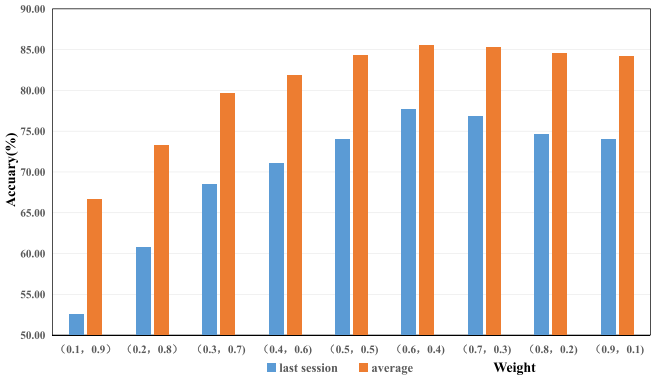


Fig. 9. Comparison of different weights for the fusion process. The accuracy of the last session and average is reported in UCMerced-LandUse.

We can see from Fig. 8 that the accuracy of the G-MFCN model in the incremental sessions roughly increases with an increasing number of shots. The growth rate of accuracy is rapid for shot numbers below five, but it slows down as the number of shots exceeds five. For FSL, it is usually considered that the categories in our dataset consist of only a few training samples, and the experiments are conducted using either three-shot or five-shot conditions to better assess the model's learning capability. Therefore, we finally choose five-shot as the test condition for our experiment.

2) Weights of the Different Networks: In this section, we establish the weights for the parallel hierarchical networks during the fusion process, denoted as (α, β). The performance metrics, specifically the accuracy of the last session and the overall average accuracy, are evaluated under various combinations of these weights. The results, as depicted in the accompanying Fig. 9, indicate that the combinations of (0.6, 0.4) and (0.7, 0.3) for the weights yield notably positive outcomes for both the last session and the average accuracy. This observation further substantiates the efficacy of the proposed G-MFCN.

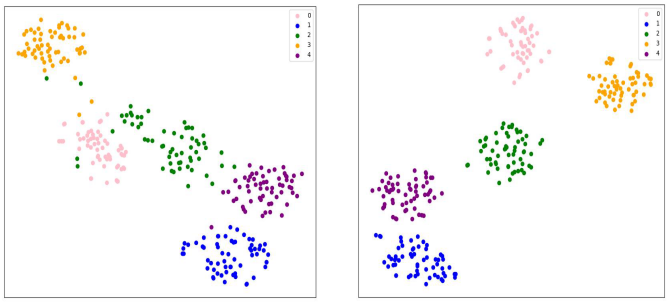
F. t-SNE Visualization

In this section, we verify the effectiveness of G-MFCN by the t-SNE visualization method. This article takes advantage

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ACCURACY OF DIFFERENT STATE-OF-THE-ART FSCIL METHODS ON THE CUB200 DATASET

Method	Session 0	Session 1	Session 2	Session 3	Session 4	Session 5	Session 6	Session 7	Session 8	Session 9	Session 10	Average	Relative Improvement
EEIL (ECCV 2018) [39]	68.68	52.65	48.61	44.16	36.62	29.52	27.83	26.26	24.01	23.89	21.16	36.67	+30.69
NCM (CVPR 2019) [22]	68.68	57.12	44.21	28.78	26.71	25.66	24.62	21.52	20.12	20.06	19.87	36.27	+31.09
TOPIC (CVPR 2020) [13]	68.68	62.49	54.81	49.99	45.25	41.40	38.35	35.36	32.22	28.31	26.28	43.92	+23.44
DEMD (CVPR 2020) [40]	75.35	70.69	66.68	62.34	59.76	56.54	54.61	52.52	50.73	49.20	47.60	58.73	+8.63
SS-NCM (ICIP 2021) [41]	69.89	61.91	55.51	51.71	49.68	46.11	42.19	39.03	37.96	34.05	32.65	47.33	+20.03
SAKD (CVPR 2021) [25]	68.23	60.45	55.70	50.45	45.72	42.90	40.89	38.77	36.51	34.87	32.96	46.13	+21.23
SPPR (CVPR 2021) [36]	68.68	61.85	57.43	52.68	50.19	46.88	44.65	43.07	40.17	39.63	37.33	49.32	+18.04
CEC (CVPR 2021) [14]	75.85	71.94	68.50	63.50	62.43	58.27	57.73	55.81	54.83	53.52	52.28	61.33	+6.03
ERL (AAAI 2021) [42]	73.52	70.12	65.12	62.01	58.56	57.99	56.77	56.52	55.01	53.68	50.01	59.93	+7.43
FSLL+SS (AAAI 2021) [43]	75.63	71.81	68.16	64.32	62.61	60.10	58.82	58.70	56.45	56.41	55.82	62.62	+4.74
F2M (NIPS 2021) [44]	77.13	73.92	70.27	66.37	64.34	61.69	60.52	59.38	57.15	56.94	55.89	63.96	+3.40
IDLVQ-C (ICLR 2021) [26]	77.37	74.72	70.28	67.13	65.34	63.52	62.10	61.54	59.04	58.68	57.81	65.23	+2.13
MetaFSCIL (CVPR 2022) [45]	75.90	72.41	68.78	64.78	62.96	59.99	58.30	56.85	54.78	53.82	52.64	61.93	+5.43
FeSSSS (CVPR 2022) [46]	79.60	73.46	70.32	66.38	63.97	59.63	58.19	57.56	55.01	54.31	52.98	62.85	+4.51
FACT (CVPR 2022) [15]	75.90	73.23	70.84	66.13	65.56	62.15	61.74	59.83	58.41	57.89	56.94	64.42	+2.94
ALICE (ECCV 2022) [47]	77.40	72.70	70.60	67.20	65.90	63.40	62.90	61.90	60.50	60.60	60.10	65.75	+1.61
Liu et al. (ECCV 2022) [48]	75.90	72.14	68.64	63.76	62.58	59.11	57.82	55.89	54.92	53.58	52.39	61.52	+5.84
LIMIT (TPAMI 2022) [38]	75.89	73.55	71.99	68.14	67.42	63.61	62.40	61.35	59.91	58.66	57.41	65.48	+1.88
SSFE-Net (WACV2023) [17]	76.38	72.11	68.82	64.77	63.59	60.56	59.84	58.93	57.33	56.23	54.28	62.99	+4.37
MCNet (TIP 2023) [49]	77.57	73.96	70.47	65.81	66.16	63.81	62.09	61.82	60.41	60.09	59.08	65.57	+1.79
SoftNet (ICLR 2023) [50]	78.07	74.58	71.37	67.54	65.37	62.60	61.07	59.37	57.53	57.21	56.75	64.68	+2.68
UaD-CIE (TNNLS 2023) [51]	75.17	73.27	70.87	67.14	65.45	63.66	62.42	62.55	60.99	60.48	60.72	65.70	+1.66
Entropy-reg (CVPR 2023) [52]	78.88	75.62	72.32	68.62	67.23	64.26	62.98	61.89	60.20	59.21	58.67	66.35	+1.00
G-MFCN with CEC	**79.99**	**76.67**	**73.68**	**68.98**	**68.26**	**64.99**	**64.12**	**62.49**	**61.25**	**60.67**	**59.90**	**67.36**	



(a) (b)

Fig. 10. t-SNE figures of features on the UC Merced dataset. (a) Features extracted by the CEC method. (b) Features extracted from the G-MFCN + CEC fusion model.

of the t-SNE diagram to visualize the features extracted by ResNet-18 and the new data features extracted by the modified multiscale fusion model G-MFCN on the UC dataset. For t-SNE, sample characteristics pertaining to the same category should be clustered together. In Fig. 10, we find that although original method (left figure) is effective for classification, the extracted features have the deficiency of distribution shift and confusion between different classes. The features on the right exhibit a greater interclass direct distance and a reduced intraclass distance, which further demonstrates that our G-MFCN can strengthen feature discrimination and enhance the performance of classification.

V. CONCLUSION

In this article, we propose a G-MFCN, which is used for improving the performance of FSCIL-RSSC. Our G-MFCN aims to concentrate on designing a robust feature backbone trained on the base classes and using the multiscale features extracted from it to enhance the discrimination of new class samples' embeddings. To obtain more comprehensive features, we design two parallel CNNs for simultaneous feature extraction, and then fuse resulting multiscale features in top–down sequence. Moreover, based on the network structure of ResNet-18, we propose a gradient guided approach to score the feature extraction performance of each layer and pick out the optimal layer for fusion. Numerous experiments on four public datasets demonstrate the superiority of G-MFCN. More importantly, the G-MFCN is a general model that can be embedded into the existing FSCIL architectures to enhance performance of the model.

As for FSCIL-RSSC, there are few studies, but many questions are worth exploring in this field. Since remote sensing images are greatly disturbed by the environment, in future work, we aim to introduce a foreground enhancement mechanism to improve the discrimination between foreground and background targets. This will enhance the accuracy and stability of classification algorithms by focusing more on the target itself. In addition, to address the challenge of the limited availability of remote sensing images, we plan to explore the combination of FSCIL with cross-domain techniques. This will enable the transfer and application of knowledge by adapting the trained feature extraction model to different types of remote sensing datasets and new scenes.

APPENDIX

Because of few studies of FSCIL-RSSC, our experiments lack comparative approaches for scene classification using remote sensing images. To further verify the validity of G-MFCN, in the supplementary section, we provide the experimental results compared with the cub200 dataset which is commonly used in the field of FSCIL. Table VI lists the experimental results of numerous state-of-the-art incremental learning models with FSL in recent years. Here, we take the classical CEC as the baseline, ResNet-18 as the backbone of the network, and other configurations align with the primary experiment (Section IV-C) in this article. In addition, the Top-1 precision for each accuracy and the average accuracy across all the sessions are used for measures. According to Table VI, it can be explicitly found that G-MFCN is superior to most of the existing advanced FSCIL approaches and can be embedded into the existing FSCIL baseline architectures to enhance model performance.

REFERENCES

[1] J. Zhang, J. Liu, B. Pan, and Z. Shi, "Domain adaptation based on correlation subspace dynamic distribution alignment for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7920–7930, Nov. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

[2] B. Zhao, Y. Zhong, G. Xia, and L. Zhang, "Dirichlet-derived multiple topic scene classification model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2108–2123, Apr. 2016.

[3] A. E. Maxwell, T. A. Warner, and F. Fang, "Implementation of machine-learning classification in remote sensing: An applied review," *Int. J. Remote Sens.*, vol. 39, no. 9, pp. 2784–2817, May 2018.

[4] P. Du, J. Xia, J. Chanussot, and X. He, "Hyperspectral remote sensing image classification based on the integration of support vector machine and random forest," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2012, pp. 174–177.

[5] C. Xu, G. Zhu, and J. Shu, "A lightweight and robust lie group-convolutional neural networks joint representation for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501415.

[6] V. S. Martins, A. L. Kaleita, B. K. Gelder, H. L. F. da Silveira, and C. A. Abe, "Exploring multiscale object-based convolutional neural network (multi-OCNN) for remote sensing image classification at high spatial resolution," *ISPRS J. Photogramm. Remote Sens.*, vol. 168, pp. 56–73, Oct. 2020.

[7] X. Lu, X. Zheng, and Y. Yuan, "Remote sensing scene classification by unsupervised representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5148–5157, Sep. 2017.

[8] S. Fu, Q. Cao, Y. Lei, Y. Zhong, Y. Zhan, and X. You, "Few-shot learning with dynamic graph structure preserving," *IEEE Trans. Ind. Informat.*, early access, Sep. 30, 2023, doi: 10.1109/TII.2023.3306929.

[9] S. Fu et al., "Adaptive multi-scale transductive information propagation for few-shot learning," *Knowl.-Based Syst.*, vol. 249, Aug. 2022, Art. no. 108979.

[10] L. Li, J. Han, X. Yao, G. Cheng, and L. Guo, "DLA-MatchNet for few-shot remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7844–7853, Sep. 2021.

[11] Z. Yuan and W. Huang, "Multi-attention DeepEMD for few-shot learning in remote sensing," in *Proc. IEEE 9th Joint Int. Inf. Technol. Artif. Intell. Conf. (ITAIC)*, vol. 9, Dec. 2020, pp. 1097–1102.

[12] A. Han, L. Xing, W. Liu, and B. Liu, "MVFF: Multi-view feature fusion for few-shot remote sensing image scene classification," in *Proc. IEEE Int. Conf. Syst., Man, Cybern. (SMC)*, Oct. 2022, pp. 2452–2458.

[13] X. Tao, X. Hong, X. Chang, S. Dong, X. Wei, and Y. Gong, "Few-shot class-incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12183–12192.

[14] C. Zhang, N. Song, G. Lin, Y. Zheng, P. Pan, and Y. Xu, "Few-shot incremental learning with continually evolved classifiers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 12455–12464.

[15] D.-W. Zhou, F.-Y. Wang, H.-J. Ye, L. Ma, S. Pu, and D.-C. Zhan, "Forward compatible few-shot class-incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 9046–9056.

[16] Z. Song, Y. Zhao, Y. Shi, P. Peng, L. Yuan, and Y. Tian, "Learning with fantasy: Semantic-aware virtual contrastive constraint for few-shot class-incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 24183–24192.

[17] Z. Pan, X. Yu, M. Zhang, and Y. Gao, "SSFE-net: Self-supervised feature enhancement for ultra-fine-grained few-shot class incremental learning," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6264–6273.

[18] W. Liu, X. Nie, B. Zhang, and X. Sun, "Incremental learning with open-set recognition for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622916.

[19] Y. Zeng, B. Xue, and W. Han, "Few-shot incremental learning for aerial image scene classification based on feature adaptation and prototype continuous optimization," in *Proc. IEEE Int. Conf. Unmanned Syst. (ICUS)*, Oct. 2022, pp. 510–515.

[20] Z. Zhu, P. Wang, W. Diao, J. Yang, H. Wang, and X. Sun, "Few-shot incremental learning with continual prototype calibration for remote sensing image fine-grained classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 196, pp. 210–227, Feb. 2023.

[21] Y. Tai, Y. Tan, S. Xiong, and J. Tian, "Mine-distill-prototypes for complete few-shot class-incremental learning in image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5206013.

[22] S. Hou, X. Pan, C. C. Loy, Z. Wang, and D. Lin, "Learning a unified classifier incrementally via rebalancing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 831–839.

[23] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 3630–3638.

[24] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, Dec. 2017.

[25] A. Cheraghian, S. Rahman, P. Fang, S. K. Roy, L. Petersson, and M. Harandi, "Semantic-aware knowledge distillation for few-shot class-incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2534–2543.

[26] J. Zhu, L. Liu, Y. Zhan, X. Zhu, H. Zeng, and D. Tao, *Int. J. Comput. Vis.*, vol. 131, no. 11, pp. 2959–2976, 2023.

[27] T. Y. Lin, P. Dollàr, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 2117–2125.

[28] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.

[29] Y. Li, Y. Chen, N. Wang, and Z.-X. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6054–6063.

[30] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 3146–3154.

[31] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5987–5995.

[32] X. Huo et al., "HiFuse: Hierarchical multi-scale feature fusion network for medical image classification," *Biomed. Signal Process. Control*, vol. 87, Jan. 2024, Art. no. 105534.

[33] W. Qiu, S. Fu, J. Zhang, C. Lei, and Q. Peng, "Semantic-visual guided transformer for few-shot class-incremental learning," in *Proc. IEEE Int. Conf. Multimedia Expo*, Mar. 2023, pp. 2885–2890.

[34] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[35] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograph. Inf. Syst.*, Nov. 2010, pp. 270–279.

[36] K. Zhu, Y. Cao, W. Zhai, J. Cheng, and Z.-J. Zha, "Self-promoted prototype refinement for few-shot class-incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 6797–6806.

[37] M. Hersche, G. Karunaratne, G. Cherubini, L. Benini, A. Sebastian, and A. Rahimi, "Constrained few-shot class-incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 9057–9067.

[38] D.-W. Zhou, H.-J. Ye, L. Ma, D. Xie, S. Pu, and D.-C. Zhan, "Few-shot class-incremental learning by sampling multi-phase tasks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 12816–12831, Nov. 2023.

[39] F. M. Castro, M. J. Marín-Jiménez, N. Guil, C. Schmid, and K. Alahari, "End-to-end incremental learning," in *Proc. ECCV*, 2018, pp. 233–248.

[40] C. Zhang, Y. Cai, G. Lin, and C. Shen, "DeepEMD: Differentiable Earth mover's distance for few-shot learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, p. 12203—12213.

[41] Y. Cui, W. Xiong, M. Tavakolian, and L. Liu, "Semi-supervised few-shot class-incremental learning," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 1239–1243.

[42] S. Dong, X. Hong, X. Tao, X. Chang, X. Wei, and Y. Gong, "Few-shot class-incremental learning via relation knowledge distillation," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 2, pp. 1255–1263.

[43] P. Mazumder, P. Singh, and P. Rai, "Few-shot lifelong learning," in *Proc. AAAI Conf. Artif. Intell.*, 2021, vol. 35, no. 3, pp. 2337–2345.

[44] G. Shi, J. Chen, W. Zhang, L. Zhan, and X. M. Wu, "Overcoming catastrophic forgetting in incremental few-shot learning by finding flat minima," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 6747–6761.

[45] Z. Chi, L. Gu, H. Liu, Y. Wang, Y. Yu, and J. Tang, "MetaFSCIL: A meta-learning approach for few-shot class incremental learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 14146–14155.

[46] T. Ahmad et al., "Few-shot class incremental learning leveraging self-supervised features," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 3899–3909.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.

[47] C. Peng, K. Zhao, T. Wang, M. Li, and B. C. Lovell, "Few-shot class-incremental learning from an open-set perspective," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 382–397.
[48] H. Liu et al., "Few-shot class-incremental learning via entropy-regularized data-free replay," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 146–162.
[49] Z. Ji, Z. Hou, X. Liu, Y. Pang, and X. Li, "Memorizing complementation network for few-shot class-incremental learning," *IEEE Trans. Image Process.*, vol. 32, pp. 937–948, 2023.
[50] J. Yoon, S. Madjid, S. J. Hwang, and C.-D. Yoo, "On the soft-subnetwork for few-shot class incremental learning," in *Proc. Int. Conf. Learn. Represent.*, 2023, pp. 1–23.
[51] Y. Cui, W. Deng, H. Chen, and L. Liu, "Uncertainty-aware distillation for semi-supervised few-shot class-incremental learning," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, May 31, 2023, doi: 10.1109/TNNLS.2023.3277018.
[52] H. Zhuang, Z. Weng, R. He, Z. Lin, and Z. Zeng, "GKEAL: Gaussian kernel embedded analytic learning for few-shot class incremental task," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7746–7755.

Wuli Wang (Member, IEEE) received the B.S. and M.S. degrees from the China University of Petroleum, Qingdao, China, in 2001 and 2007, respectively, and the Ph.D. degree from Chongqing University, Chongqing, China, in 2018.

He is currently an Associate Professor with the College of Oceanography and Space Informatics, China University of Petroleum. His research interests include machine learning and remote sensing image processing.

Li Zhang received the B.E. degree in electronic information science and technology from Shandong Agricultural University, Tai'an, China, in 2022. She is currently pursuing the M.S. degree in information and communication engineering with the China University of Petroleum (East China), Qingdao, China.

Her research interests include deep learning, class-incremental learning, and remote sensing scene classification.

Sichao Fu (Graduate Student Member, IEEE) is currently pursuing the Ph.D. degree with the School of Electronic Information and Communications, Huazhong University of Science and Technology, Wuhan, China.

His research interests include pattern recognition and deep manifold learning.

Peng Ren (Senior Member, IEEE) received the B.Eng. and M.Eng. degrees in electronic engineering from the Harbin Institute of Technology, Harbin, China, in 2003 and 2005, respectively, and the Ph.D. degree in computer science from the University of York, York, U.K., in 2010.

He is currently a Professor with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao, China. His research interests include remote sensing and machine learning.

Guangbo Ren received the Ph.D. degree in marine information detection and processing from the Ocean University of China, Qingdao, China, in 2010.

He is currently an Associate Professor with the First Institute of Oceanography, Ministry of Natural Resources, Qingdao. His research interests include coastal ecosystem remote sensing technology study and application.

Qinmu Peng received the Ph.D. degree from the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China, in 2015.

He is currently an Associate Professor with the School of Electronic Information and Communications, Huazhong University of Science and Technology, Wuhan, China. His research interests include medical image processing, pattern recognition, machine learning, and computer vision.

Baodi Liu (Member, IEEE) received the Ph.D. degree in electronic engineering from Tsinghua University, Beijing, China, in 2013.

He is currently an Associate Professor with the College of Control Science and Engineering, China University of Petroleum, Qingdao, China. His research interests include computer vision and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:36:40 UTC from IEEE Xplore. Restrictions apply.